

Calculus A

Chapter 2: Limits and Continuity

Section 2.2: Limit of a Function and Limit Laws

Main Lecture Notes

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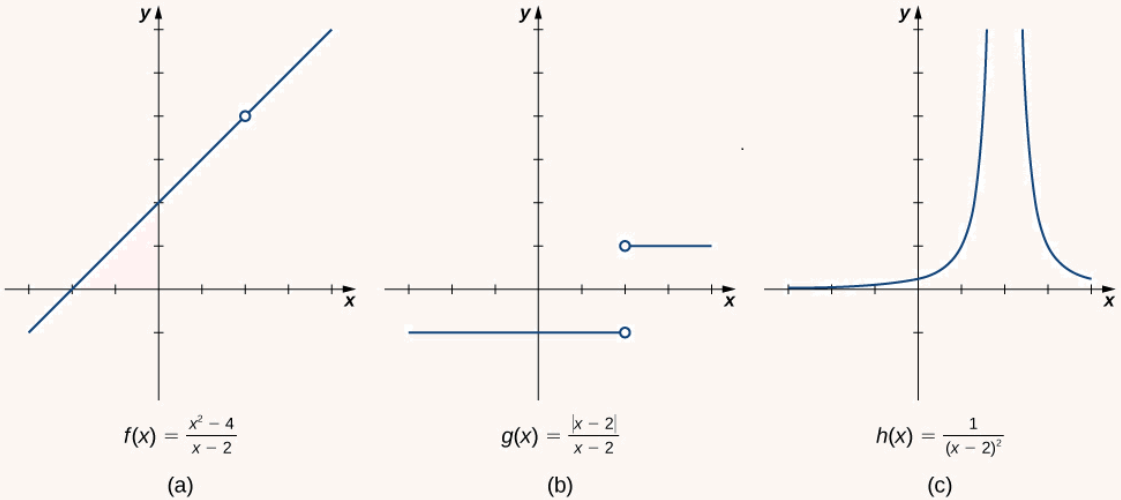
Reference: Thomas' Calculus, 12th Edition

Exploring the Limit Concept

We begin our exploration of limits by taking a look at the graphs of the functions

$$f(x) = \frac{x^2 - 4}{x - 2} \qquad g(x) = \frac{|x - 2|}{x - 2} \text{ and, } \qquad h(x) = \frac{1}{(x - 2)^2}$$

which are shown in the Figure below. In particular, let's focus our attention on the behavior of each graph at and around $x = 2$.



Each of the three functions is undefined at $x = 2$, but if we make this statement and no other, we give a very incomplete picture of how each function behaves in the neighborhood of $x = 2$. To express the behavior of each graph in the neighborhood of 2 more completely, we need to introduce the concept of a limit.

Note

Let's first take a closer look at how the function $f(x) = \frac{x^2-4}{x-2}$ behaves around $x = 2$ in part (a) of the Figure above. As the values of x approach 2 from either side of 2, the values of $y = f(x)$ approach 4. Mathematically, we say that the limit of $f(x)$ as x approaches 2 is 4. Symbolically, we express this limit as

$$\lim_{x \to 2} f(x) = 4.$$

From this very brief informal look at one limit, let's start to develop an informal definition of the limit. We can think of the limit of a function at a number x_0 as being the one real number L that the functional values approach as the $x-$values approach x_0, provided such a real number L exists. Stated more carefully, we have the following definition:

Informal Definition of the Limit

Let $f(x)$ be a function defined at all values in an open interval containing x_0, with the possible exception of a itself, and let L be a real number. If all values of the function $f(x)$ approach the real number L as the values of $x(\neq x_0)$ approach the number x_0, then we say that the limit of $f(x)$ as x approaches x_0 is L.

(In other words, as x gets closer to x_0, $f(x)$ gets closer and stays close to L.) Symbolically, we express this idea as

$$\lim_{x \to x_0} f(x) = L.$$

Example 1

Evaluate $\lim_{x\to 1} f(x) = \lim_{x\to 1} \dfrac{x(x-1)}{x-1}$.

Solution:

From above, we see that

$$\frac{x(x-1)}{x-1} = x \qquad \text{if } x \neq 1.$$

So $\lim_{x\to 1} f(x) = \lim_{x\to 1} \dfrac{x(x-1)}{x-1} = \lim_{x\to 1} x$ if $x \neq 1$. Therefore, in words, we ask "what does x get close to when x is close to 1?"
Well ... x gets close to 1! So

$$\lim_{x\to 1} f(x) = \lim_{x\to 1} \frac{x(x-1)}{x-1} = \lim_{x\to 1} x = 1,$$

even though we are not allowing x to equal 1; the important thing is that x can be made arbitrarily close to 1.

Example 2: Evaluating a Limit Using a Table of Functional Values

Evaluate $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$ using a table of functional values.

Solution:

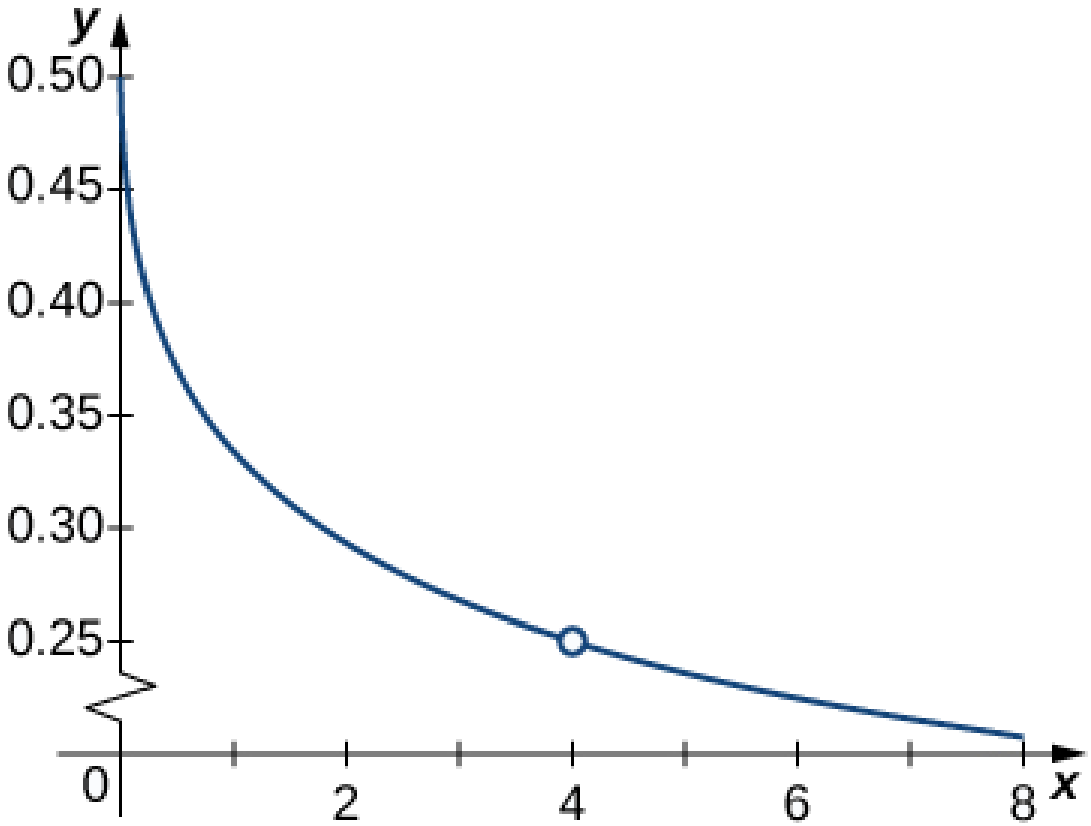
The values of $f(x) = \frac{\sqrt{x}-2}{x-4}$ for the values of x listed in Table below:

x	$\frac{\sqrt{x}-2}{x-4}$	x	$\frac{\sqrt{x}-2}{x-4}$
3.9	0.251582341869	4.1	0.248456731317
3.99	0.25015644562	4.01	0.24984394501
3.999	0.250015627	4.001	0.249984377
3.9999	0.250001563	4.0001	0.249998438
3.99999	0.25000016	4.00001	0.24999984

After inspecting this table, we see that the functional values less than 4 appear to be decreasing toward 0.25 whereas the functional values greater than 4 appear to be increasing toward 0.25. We conclude that $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4} = 0.25$.

Example 2 Cont..

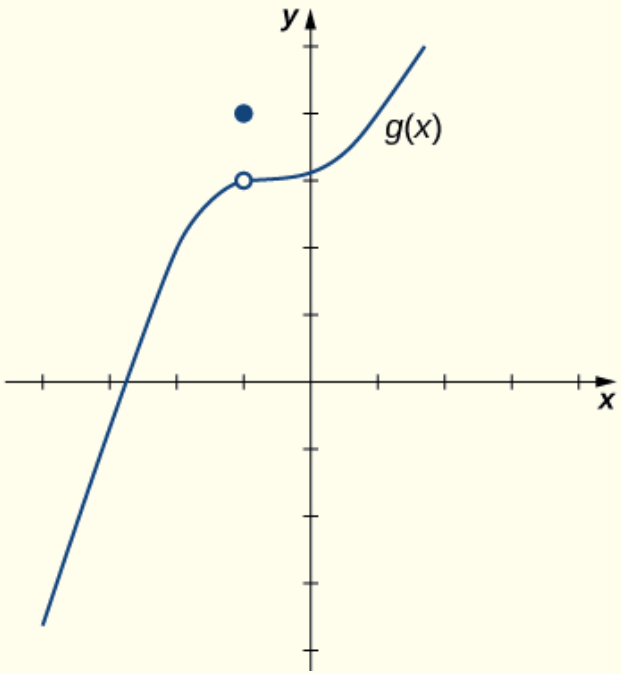
We confirm this estimate using the graph of $f(x) = \frac{\sqrt{x}-2}{x-4}$ shown in the Figure below:



Example 3: Evaluating a Limit Using a Graph

For $g(x)$ shown in the Figure below, evaluate $\lim_{x \to -1} g(x)$.

Solution:



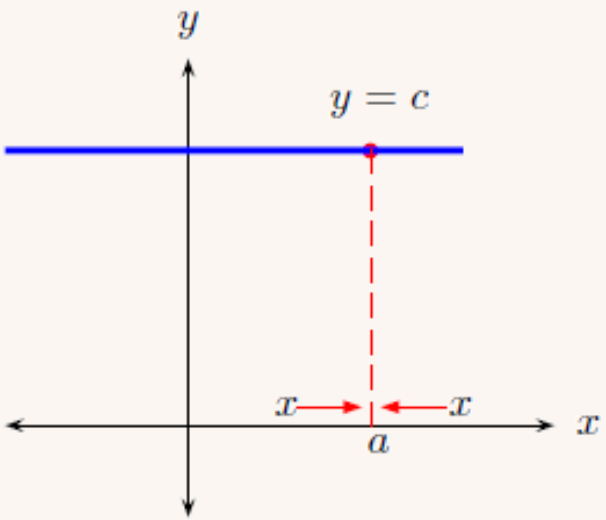
Despite the fact that $g(-1) = 4$, as the $x-$values approach -1 from either side, the $g(x)$ values approach 3. Therefore, $\lim_{x \to -1} g(x)$. Note that we can determine this limit without even knowing the algebraic expression of the function.

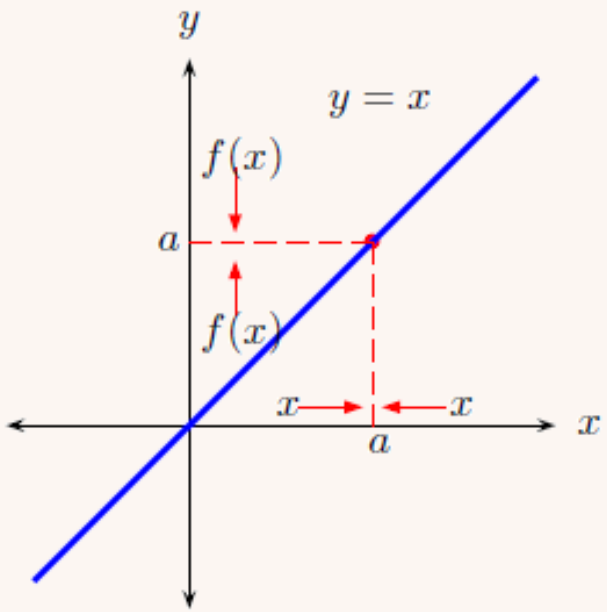
Evaluating Limits with the Limit Laws

Two Important Limits

Let a be a real number and c be a constant.

1. Identity Law: $\lim_{x \to a} x = a$
2. Constant Law: $\lim_{x \to a} c = c$





Two Important Limits Cont..

We can make the following observations about these two limits.

1. For the first limit, observe that as x approaches a, so does $f(x)$, because $f(x) = x$. Consequently, $\lim_{x \to a} x = a$.

2. For the second limit, regardless of whether they are approaching a, the values $f(x)$ remain constant at c. We have no choice but to conclude $\lim_{x \to a} c = c$.

Evaluating Limits with the Limit Laws

Limit Laws

Let $f(x)$ and $g(x)$ be defined for all $x \neq a$ over some open interval containing a. Assume that L and M are real numbers such that $\lim\limits_{x\to a} f(x) = L$ and $\lim\limits_{x\to a} f(x) = L$. Let c be a constant. Then, each of the following statements holds:

Sum law: $\lim\limits_{x\to a}(f(x)+g(x)) = \lim\limits_{x\to a} f(x) + \lim\limits_{x\to a} g(x) = L+M$

Difference law: $\lim\limits_{x\to a}(f(x)-g(x)) = \lim\limits_{x\to a} f(x) - \lim\limits_{x\to a} g(x) = L-M$

Constant multiple law: $\lim\limits_{x\to a} c.f(x) = c.\lim\limits_{x\to a} f(x) = c.L$

Product law: $\lim\limits_{x\to a}(f(x).g(x)) = \lim\limits_{x\to a} f(x).\lim\limits_{x\to a} g(x) = L.M$

Quotient law: $\lim\limits_{x\to a}\dfrac{f(x)}{g(x)} = \dfrac{\lim\limits_{x\to a} f(x)}{\lim\limits_{x\to a} g(x)} = \dfrac{L}{M}, \quad \text{for } M \neq 0$

Power law: $\lim\limits_{x\to a}(f(x))^n = (\lim\limits_{x\to a} f(x))^n = L^n \qquad n > 0$

Root law: $\lim\limits_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{\lim\limits_{x\to a} f(x)} = \sqrt[n]{L}$

for all L if n is odd and for $L \geq 0$ if n is even.

Evaluating a Limit Using Limit Laws

Example 1

Use the limit laws to evaluate

1. $\lim_{x\to -3}(4x+2)$.
2. $\lim_{x\to 2}\dfrac{2x^2-3x+1}{x^3+4}$.
3. $\lim_{x\to 6}(2x-1)\sqrt{x+4}$.

Solution: 1

$$\begin{aligned}\lim_{x\to -3}(4x+2) &= \lim_{x\to -3}4x+\lim_{x\to -3}2 \quad \text{Use the sum law.}\\ &= 4.\lim_{x\to -3}x+\lim_{x\to -3}2 \quad \text{Use constant multiple law.}\\ &= 4.(-3)+2=-10.\end{aligned}$$

Evaluating a Limit Using Limit Laws

Example 1 Cont..

$$\lim_{x\to 2}\frac{2x^2-3x+1}{x^3+4}.$$

Solution: 2

$$\lim_{x\to 2}\frac{2x^2-3x+1}{x^3+4}=\frac{\lim_{x\to 2}(2x^2-3x+1)}{\lim_{x\to 2}(x^3+4)}$$ Use the quotient law.

$$=\frac{2.\lim_{x\to 2}x^2-3.\lim_{x\to 2}x+\lim_{x\to 2}1}{\lim_{x\to 2}x^3+\lim_{x\to 2}4}$$ sum law & constant multiple.

$$=\frac{2.(\lim_{x\to 2}x)^2-3.\lim_{x\to 2}x+\lim_{x\to 2}1}{(\lim_{x\to 2}x)^3+\lim_{x\to 2}4}$$ Use the power law.

$$=\frac{2(4)-3(2)+1}{(2)^3+4}=\frac{1}{4}.$$

Evaluating a Limit Using Limit Laws

Example 1 Cont..

$\lim_{x\to 6}(2x-1)\sqrt{x+4}.$

Solution: 3

$$\lim_{x\to 6}(2x-1)\sqrt{x+4} = \lim_{x\to 6}(2x-1).\lim_{x\to 6}\sqrt{x+4} \quad \text{Use the product law.}$$
$$= 11\sqrt{10}$$

Limits of Polynomial and Rational Functions

Let $p(x)$ and $q(x)$ be polynomial functions. Let a be a real number. Then,

$$\lim_{x\to a} p(x) = p(a)$$

$$\lim_{x\to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}$$

Example 2

Find the following limits

1. $\lim\limits_{x\to 3} \dfrac{2x^2-3x+1}{5x+4}$.
2. $\lim\limits_{x\to -2}(3x^3-2x+7)$.

Solution:

1. $$\lim_{x\to 3}\frac{2x^2-3x+1}{5x+4}=\frac{\lim_{x\to 3}(2x^2-3x+1)}{\lim_{x\to 3}(5x+4)}$$ Use the quotient law.
$$=\frac{2.\lim_{x\to 3}x^2-3.\lim_{x\to 3}x+\lim_{x\to 3}1}{5.\lim_{x\to 3}x+\lim_{x\to 3}4}$$
$$=\frac{10}{19}.$$ Use sum law & constant multiple.

2. $$\lim_{x\to -2}(3x^3-2x+7)=-13$$

Additional Limit Evaluation Techniques

Evaluating a Limit by Factoring and Canceling – Example 1

Evaluate: $\lim_{x\to 3} \dfrac{x^2-3x}{2x^2-5x-3}$.

Solution:

Step 1. The function $f(x) = \lim_{x\to 3} \dfrac{x^2-3x}{2x^2-5x-3}$ is undefined for $x = 3$. In fact, if we substitute 3 into the function we get $\left(\frac{0}{0}\right)$, which is undefined. Factoring and canceling is a good idea:

$$\lim_{x\to 3} \frac{x^2-3x}{2x^2-5x-3} = \lim_{x\to 3} \frac{x(x-3)}{(x-3)(2x+1)}$$

Step 2. For all $x \neq 3$, $\dfrac{x^2-3x}{2x^2-5x-3} = \dfrac{x}{(2x+1)}$. Therefore,

$$\lim_{x\to 3} \frac{x^2-3x}{2x^2-5x-3} = \lim_{x\to 3} \frac{x}{(2x+1)}.$$

Step 3. Evaluate using the limit laws:

$$\lim_{x\to 3} \frac{x}{(2x+1)} = \frac{3}{7}.$$

Evaluating a Limit by Factoring and Canceling – Example 2

Evaluate: $\lim_{x \to -3} \dfrac{x^2 + 4x + 3}{x^2 - 9}$

Solution:

$$\lim_{x \to -3} \frac{x^2 + 4x + 3}{x^2 - 9} = \frac{-2}{-6} = \frac{1}{3}. \quad (\textit{Check!!!})$$

Additional Limit Evaluation Techniques

Evaluating a Limit by Multiplying by a Conjugate – Example 1

Evaluate: $\lim_{x\to -1} \dfrac{\sqrt{x+2}-1}{x+1}$.

Solution:

Step 1. $\dfrac{\sqrt{x+2}-1}{x+1}$. has the form $\left(\frac{0}{0}\right)$, at -1. Let's begin by multiplying by $\sqrt{x+2}+1$, the conjugate of $\sqrt{x+2}-1$, on the numerator and denominator:

$$\lim_{x\to -1} \frac{\sqrt{x+2}-1}{x+1} = \lim_{x\to -1} \frac{\sqrt{x+2}-1}{x+1}.\frac{\sqrt{x+2}+1}{\sqrt{x+2}+1}.$$

Step 2. We then multiply out the numerator. We don't multiply out the denominator because we are hoping that the $(x+1)$ in the denominator cancels out in the end:

$$= \lim_{x\to -1} \frac{x+1}{(x+1)(\sqrt{x+2}+1)}.$$

Solution Cont..

Step 3. Then we cancel:

$$= \lim_{x \to -1} \frac{1}{\sqrt{x+2}+1}.$$

Step 4. Last, we apply the limit laws:

$$\lim_{x \to -1} \frac{1}{\sqrt{x+2}+1} = \frac{1}{2}.$$

Additional Limit Evaluation Techniques

Evaluating a Limit by Multiplying by a Conjugate – Example 2

Evaluate: $\lim\limits_{x\to 5} \dfrac{\sqrt{x-1}-2}{x-5}$.

Solution:

$$\lim_{x\to 5} \frac{\sqrt{x-1}-2}{x-5} = \frac{1}{4}. \quad (\textit{Check!!!})$$

Additional Limit Evaluation Techniques

Evaluating a Limit by Simplifying a Complex Fraction – Example 1

Evaluate: $\lim\limits_{x\to 1} \dfrac{\dfrac{1}{x+1}-\dfrac{1}{2}}{x-1}$.

Solution:

Step 1. $\dfrac{\dfrac{1}{x+1}-\dfrac{1}{2}}{x-1}$ has the form $\left(\frac{0}{0}\right)$, at 1. We simplify the algebraic fraction by multiplying by $\frac{2(x+1)}{2(x+1)}$: Next, we multiply through the numerators.

$$\lim_{x\to 1} \frac{\dfrac{1}{x+1}-\dfrac{1}{2}}{x-1} = \lim_{x\to 1} \frac{\left(\dfrac{1}{x+1}-\dfrac{1}{2}\right)}{x-1}.\frac{2(x+1)}{2(x+1)} = \lim_{x\to 1} \frac{2-(x+1)}{2(x-1)(x+1)}.$$

Step 2. Do not multiply the denominators because we want to be able to cancel the factor $(x-1)$. Then, we simplify the numerator. Then, we cancel the common factors of $(x-1)$:

$$= \lim_{x\to 1} \frac{-x+1}{2(x-1)(x+1)} = \lim_{x\to 1} \frac{-(x-1)}{2(x-1)(x+1)} = \lim_{x\to 1} \frac{-1}{2(x+1)} = -\frac{1}{4}.$$

Evaluating a Limit by Simplifying a Complex Fraction – Example 2

Evaluate: $\lim\limits_{x\to -3} \dfrac{\dfrac{1}{x+2}+1}{x+3}$.

Solution:

$$\lim_{x\to -3} \frac{\dfrac{1}{x+2}+1}{x+3} = -1. \quad (\textit{Check!!!})$$

Additional Limit Evaluation Techniques

Evaluating a Limit When the Limit Laws Do Not Apply – Example 1

Evaluate: $\lim_{x\to 0}\left(\frac{1}{x}+\frac{5}{x(x-5)}\right)$.

Solution:

Both $\frac{1}{x}$ and $\frac{5}{x(x-5)}$ fail to have a limit at zero. Since neither of the two functions has a limit at zero, we cannot apply the sum law for limits; we must use a different method. In this case, we find the limit by performing addition and then applying one of our previous methods. Observe that

$$\frac{1}{x}+\frac{5}{x(x-5)}=\frac{x-5+5}{x(x-5)}=\frac{x}{x(x-5)}.$$

Thus,

$$\lim_{x\to 0}\left(\frac{1}{x}+\frac{5}{x(x-5)}\right)=\lim_{x\to 0}\frac{x}{x(x-5)}=\lim_{x\to 0}\frac{1}{x-5}=-\frac{1}{5}.$$

Evaluating a Limit When the Limit Laws Do Not Apply – Example 2

Evaluate: $\lim_{x\to 3}\left(\frac{1}{x-3}-\frac{4}{x^2-2x-3}\right)$.

Solution:

Use the same method as in Example1 above. Don't forget to factor x^2-2x-3 before getting a common denominator.

$$\lim_{x\to -3}\lim_{x\to 3}\left(\frac{1}{x-3}-\frac{4}{x^2-2x-3}\right)=-\frac{1}{4}. \quad (\textit{Check!!!})$$

Theorem: The Sandwich Theorem

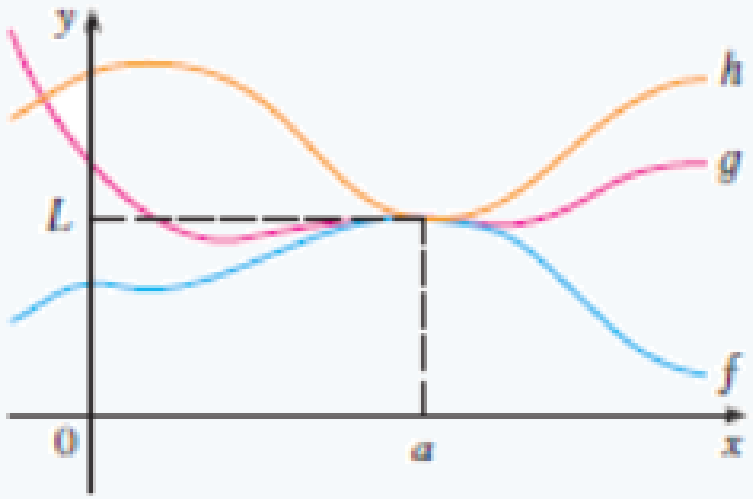
Let $f(x), g(x)$, and $h(x)$ be defined for all $x \neq a$ over an open interval containing a. If

$$f(x) \leq g(x) \leq h(x)$$

for all $x \neq a$ in an open interval containing a and

$$\lim_{x \to a} f(x) = L = \lim_{x \to a} h(x)$$

where L is a real number, then $\lim_{x \to a} g(x) = L$.



Sandwich Theorem

Example 1

If $1 - \dfrac{x^2}{4} \leq u(x) \leq 1 + \dfrac{x^2}{2}$, find $\lim\limits_{x \to 0} u(x)$.

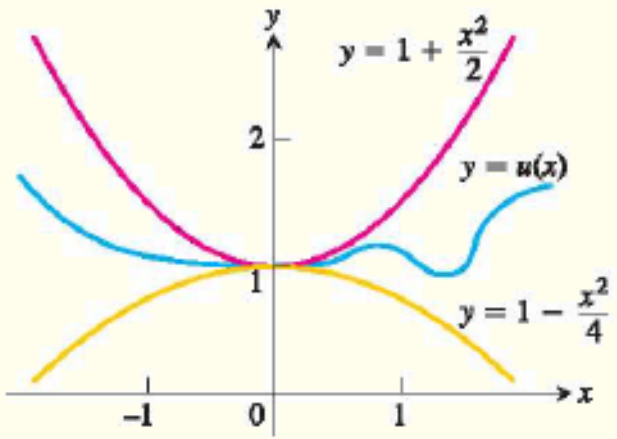
Solution:

Since

$$\lim_{x \to 0} 1 - \frac{x^2}{4} = 1 - 0 = 1 = \lim_{x \to 0} 1 + \frac{x^2}{2},$$

then by The Sandwich Theorem we have

$$\lim_{x \to 0} u(x).$$



Sandwich Theorem

Example 2

Find $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right)$

Solution:

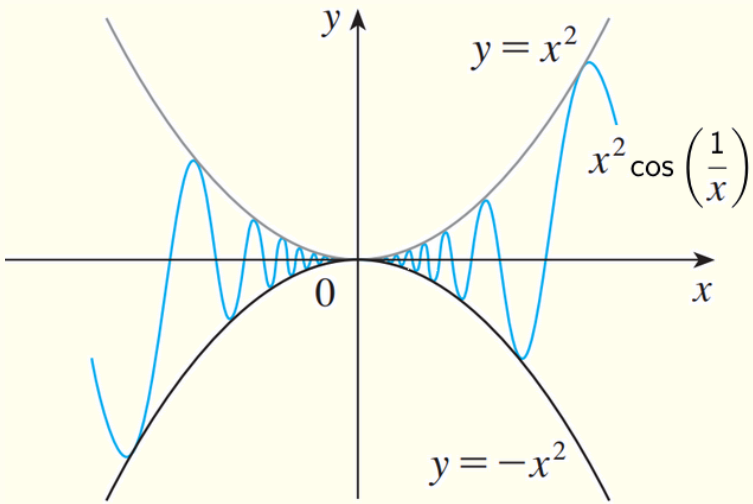
Since $-1 \le \cos\left(\frac{1}{x}\right) \le 1$ for all x

(actually we are interested only in x near 0)

then $-x^2 \le x^2 \cos\left(\frac{1}{x}\right) \le x^2$. Since

$\lim_{x\to 0} x^2 = \lim_{x\to 0} -x^2 = 0$ then by Sandwich

theorem $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right) = 0.$



End of Section 2.2.

Thank you.!!!